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Incorporation of Ammonia Borane Groups in Lithium Borohydide Structure Enables Ultrafast Lithium Ion Conductivity at Room Temperature for Solid State Batteries

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ABSTRACT: Conductor with fast Li ion conductivity at room temperature is a major challenge for the development of all solid-state Li-ion batteries. Herein, we demonstrate a novel room-temperature ultrafast Li-ion conductor, lithium borohydride ammonia borane complexes ($(LiBH_4)_x$·AB). The incorporation of AB into $LiBH_4$ structure intrinsically increases the cell volume and decreases the volume density of Li ion, which substantially facilitates the Li-ion conduction. The $LiBH_4$·AB complex delivers up to 4.04×10^{-4} S cm^{-1} of ion conductivity at 25 °C with nearly negligible electron conductivity. The Li-ion transference number is higher than 0.999 at 40 °C. The room-temperature Li-ion conductivity obtained is largely superior to other $LiBH_4$-based solid-state electrolytes reported previously under identical conditions. Moreover, the ultrafast Li-ion conduction remains stable upon heating and cooling cycling (18 – 55 ºC). Ab initio molecular dynamics simulations display a kind of 1D Li diffusion channel along the *b* direction in the $LiBH_4$·AB structure, which offers a much low activation energy barrier (0.12 eV), consequently enabling superior Li-ion conductivity at room temperature.

INTRODUCTION

Nowadays, safety problem is still a key challenge for traditional Li-ion batteries because of the highly flammable organic liquid electrolytes.[1-4] Fabricating all-solid-state batteries (ASSBs) has been considered as the ultimate solution to this issue.[5-8] By replacing organic liquid electrolytes with inorganic solid-state electrolytes (SSEs), several essential advantages can be achieved. First, the single Li-ion conducting nature and thermal stability enables ASSBs to run at a high current for fast charging/discharging. The solid state and electrochemical stability enlarges the selectivity of electrode materials. Moreover the non-fluidity makes direct stacking of cell units possible to reduce the inactive volume of a battery package to achieve a high module energy density.[9-12] A variety of SSE materials have been explored and developed for their superionic conductivity, and investigation has been extended from oxide type compounds to sulfide-based systems and metal halides, and even to complex hydrides recently.[13-16]

In 2007, Orimo and co-workers first observed the superionic conductivity of $LiBH_4$ of 10^{-2} S cm^{-1} at 170 ºC.[17] Unfortunately, the ion conductivity of approximately 10^{-8} S cm^{-1} at 30 ºC is quite low. A promising strategy towards fast room-temperature ion conductivity is to replace a fraction of the $[BH_4]^-$ anions by halide anions (I^-, Br^- or Cl^-), which stabilizes the high temperature phase of $LiBH_4$ at lower temperatures. As a representative example, the ion conductivity of $0.75LiBH_4$-$0.25LiI$ was increased by three orders of magnitude at room temperature (~ 10^{-5} S cm^{-1}) with respect to pristine $LiBH_4$.[18,19] By forming $(LiBH_4)_{1-x}(NH_2)_x$ complexes, the ion conductivity was largely enhanced to 10^{-4} - 10^{-3} S cm^{-1} at 40 ºC.[20,21] The room-temperature ion conductivity was also remarkably improved by confining $LiBH_4$ and LiI-doped $LiBH_4$ in the porous matrixes.[22,23] More encouragingly, high ion conductivity could be achieved by controlling the NH_3 content in $Li(NH_3)_nBH_4$ ($0 < n \leq 2$). For example, an ion conductivity of 2.21 × 10^{-3} S cm^{-1} was observed for $Li(NH_3)BH_4$ at 40 ºC because of the structural change caused by the absorption/desorption of NH_3,[24] especially the induced defects and/or changes in the atomic arrangement, which reduces the activation energy for mobility of Li ions by decreasing the volume density of Li ions. Note that the molecular volume of AB (V = 69.86 $Å^3$ per cell) is much larger than that of NH_3 (V = 37.79 $Å^3$ per cell), and the incorporation of AB into $LiBH_4$ structure will cause much larger volume expansion. As reported previously, the cell volume of $(LiBH_4)_2 \cdot AB$ was determined to be 681.203 $Å^3$, three times more than that of $LiBH_4$ (213.519 $Å^3$).[25,26] We therefore expect a much higher ion conductivity up to 10^{-4} - 10^{-3} S cm^{-1} at room temperature (25 ºC) by forming lithium borohydride ammonia borane complexes ($(LiBH_4)_x \cdot AB$ for short).

In this work, we conducted a first attempt to experimentally and theoretically evaluate the Li superionic conductivities of two lithium borohydride ammonia borane complexes, $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$. Ultrafast Li-ion conduction capability up to 4.04 × 10^{-4} S cm^{-1} was observed for $LiBH_4 \cdot AB$ at 25 °C, which is largely superior to other $LiBH_4$-based solid-state electrolytes reported previously under identical conditions. The Li-ion transference number was higher than 0.999 at 40 °C, with nearly negligible electronic conductivity. Theoretical simulations revealed that Li ions diffuse in a 1D channel along *b* direction in the $LiBH_4 \cdot AB$ structure with a very low energy barrier (0.12 eV), which is responsible for the superior ion conductivity. These findings provide new insights into the design and development of novel complex hydrides-based SSEs for ultrafast Li ionic conduction at room temperature.

EXPERIMENTAL SECTIONS

Materials Synthesis. $LiBH_4$ (95%, Acros) and AB (97%, Aladdin) were used as received without further purification. Two mixtures with compositions of $2LiBH_4$-AB and $LiBH_4$-AB were mechanically milled at 200 rpm for 3 h with a planetary ball mill (QM-1SP4, Nanjing). For each experiment, 1 g of the mixture was loaded into a custom-designed milling jar in a glove box (MBRAUN, Germany) filled with argon (H_2O < 0.1 ppm and O_2 < 0.1 ppm). After milling treatment, the mixtures were collected in an Ar-filled glovebox for further structural and property characterizations. For comparison, five more mixtures of $LiBH_4$-AB with molar ratios of 3:1, 2.5:1, 1.5:1, 1:1.5 and 1:2 were also prepared under identical conditions.

Materials Characterization. XRD analyses were performed on a MiniFlex 600 X-ray diffractometer (Rigaku, Japan) with Cu Kα radiation operating at 40 kV and 15 mA. Data were collected in the scattering angle (2θ) range of 10-50° with 0.02° increments. FTIR spectra were obtained with a Bruker Tensor 27 unit (Germany) in the transmission mode. The sample was mixed with potassium bromide (KBr) in 1:100 weight ratio, and then cold pressed to form a pellet. Differential scanning calorimetry (DSC) was conducted on a Netzsch DSC 200 F3 unit. Approximately 3 mg of sample was heated in an Al_2O_3 crucible from room temperature to 95 ºC at 2 ºC min^{-1}. Magic angle spinning (MAS) experiments were conducted on a Bruker Avance III 600 solid-state spectrometer using 3.2 mm double-resonance probes at a spinning rate of 15 kHz. The spectra were collected at Larmor frequencies 192.4 MHz for ^{11}B and 233.1 MHz for ^{7}Li. The ^{11}B spectra are referenced to 0.1 M H_3BO_3 solution at 19.3 ppm, Li spectra are referenced to 1 M LiCl at 0 ppm. Li spectra were collected via direct excitation at the radio frequency (RF) field of 83.3 kHz, with a relaxation delay time of 5 s. Hahn echo pulse sequence was used to acquire ^{11}B spectra.

Cells Assembly and Electrochemical Measurements. All the cells were fabricated using homemade test cells in which the battery components were placed into a PolyEtherEtherKetone (PEEK) cylinder and pressed in between two stainless steel pistons serving as electrical contacts.

In all cases, the SE powders were inserted in the cell and pressed against lithium metal or stainless-steel under 200 MPa that produce the electrolyte pellets with a diameter of 10 mm and a thickness of 2 mm.

Electrochemical impedance spectroscopy (EIS) measurements were conducted on Ivium Vertex electrochemical workstation (the Netherlands) with frequency ranging from 1 MHz to 1 Hz. Stainless steel was used on both sides of the pellets as electrodes. Before each measurement, a 40 min dwell time was used for the cell temperature to equilibrate. Each Nyquist plot consists of a depressed semicircle and a linear tail or just a linear tail, and the intersection with the Z′ axis corresponds to the resistance (*R*). The conductivity can be calculated by the following equation.

$$\sigma = \frac{d}{AR} \qquad (1)$$

where *d* is the thickness and *A* is the area of the electrolyte pellet.

The electrochemical stability window was determined via cyclic voltammetry (CV) and linear sweep voltammogram (LSV) with a scan rate of 10 μV s^{-1} in Li|$(LiBH_4)_x \cdot AB$|$(LiBH_4 \cdot AB)_x$@C|SUS cells. DC polarization measurements were carried out to determine the electronic conductivity and lithium transference number (t_+) with the SUS|$(LiBH_4)_x \cdot AB$|SUS and Li|$(LiBH_4)_x \cdot AB$|Li cells on the same electrochemical workstation. A voltage of 0.05 V was applied across the SE during the measurement. Galvanostatic platting-stripping cycles were performed using cells with lithium foil on both side as electrodes on a Neware battery test system (Shenzhen, China).

Theoretical Calculations. Ab initio molecular dynamics (AIMD) was performed to get further insight into the mechanism of Li diffusion improvement caused by structure tuning. During the simulations, density function theory calculations were performed by using the CP2K package.[27] PBE functional[28] with Grimme D3 correction[29] was used to describe the system. Unrestricted

Kohn-Sham DFT was used as the electronic structure method in the framework of the Gaussian and plane waves method.[30,31] The Goedecker-Teter-Hutter (GTH) pseudopotentials,[32,33] DZVP-MOLOPT-GTH basis sets were utilized to describe the molecules. A plane-wave energy cut-off of 500 Ry has been employed. Broyden-Fletcher-Goldfarb-Shanno algorithm (BFGS) method was used for the geometry optimization and all atoms were fully relaxed until the total energy change between two ionic steps is smaller than 10^{-7} eV. The Gamma k-point sampling was used for the Brillouin zone integration. The climbing image nudged elastic band (CI-NEB) method has been used to compute obtain the minimum energy path and the activation barrier.[34] To speed up diffusion and shorten the simulation time scale, the NVT ensemble has been performed at 900 K using Nose-Hoover chain thermostat[35,36] with the time step of 0.5 fs. The simulations are carried out in three-dimensional periodic boundary boxes of 16.62 × 12.43 × 13.19 $Å^3$ for $(LiBH_4)_2{\cdot}AB$ bulk and 14.31 × 13.09 × 15.35 $Å^3$ for $LiBH_4{\cdot}AB$ bulk, respectively. For the sake of structure stabilities, time lengths of 2.5 ps and 10 ps were used to analyze the Li-ion trajectories.

RESULTS AND DISCUSSION

Preparation and Structural Characterization of $(LiBH_4)_2{\cdot}AB$ and $LiBH_4{\cdot}AB$. Two lithium borohydride ammonia borane complexes, $(LiBH_4)_2{\cdot}AB$ and $LiBH_4{\cdot}AB$, were prepared by ball milling the corresponding mixtures of $LiBH_4$ and AB at given molar ratios. Unlike white raw materials, the resultant products were gray in color and slightly darker with increased AB content (**Figure 1**a). The X-ray diffraction (XRD) results (**Figure 1**b) indicated the formation of $(LiBH_4)_2{\cdot}AB$ and $LiBH_4{\cdot}AB$. Additionally, a little bit of impurity phases was also identified, including $LiBH_4{\cdot}AB$ in the $x = 2$ sample and AB in the $x = 1$ sample. Extending the molar ratio of $LiBH_4$-AB to 3:1 or 1:2 did not induce the formation of other new phases (**Figure S1**, Supporting Information). FTIR spectra (**Figure 1**c) revealed a good structural integrity of BH_4 and AB because the typical vibrations of B-H in BH_4 group and N-H, B-N and B-H in AB molecules remain nearly unchanged. This observation was further confirmed by solid-state nuclear magnetic resonance (NMR) characterization (**Figure 1**d) as the ^{11}B peaks assignable to $LiBH_4$ and AB were simultaneously detected at -40.41 and -25.81 ppm, respectively. This is in good agreement with their structural features. As reported previously,[25,26] the essential structural configurations of BH_4 and AB groups, whether present in $(LiBH_4)_2 \cdot AB$ or in $LiBH_4 \cdot AB$, were identical to the parent $LiBH_4$ and AB, although $LiBH_4$ components exist a layered arrangement in $(LiBH_4)_2 \cdot AB$ while they gather into columns running along the *b* direction in $LiBH_4 \cdot AB$ (**Figure S2**, Supporting Information). Compared with $(LiBH_4)_2 \cdot AB$, moreover, the intensities of N-H, B-N and B-H vibrations belonging to AB groups intensified in $LiBH_4 \cdot AB$ (**Figure 1**c). Similar phenomenon was also observed in the ^{11}B solid-state NMR spectra. This is attributed to the increased relative content of AB in $LiBH_4 \cdot AB$ with respect to $(LiBH_4)_2 \cdot AB$.

Li Ion Conductivity of $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$. The prepared $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$ samples were assembled into symmetric cells consisting of SUS|$(LiBH_4)_x \cdot AB$|SUS (SUS: stainless-steel) to assess their ion conductivities. As a reference, the ion conductivity of $LiBH_4$ was also estimated under identical conditions. **Figure 2**a-c shows the electrochemical impedance spectra (EIS) obtained from $LiBH_4$, $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$ in the frequency range of 1 MHz – 1 Hz at 25 ºC, respectively. Whereas the EIS plot of pristine $LiBH_4$ displays only a single semicircle, the EIS plots of $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$ are composed of a semicircle in the high frequency region and a sloping line in low frequency region. In addition, the diameters of the high-frequency semicircles decreased dramatically with the incorporation of

AB into $LiBH_4$ structure, and even became almost undetectable for the $LiBH_4 \cdot AB$, implying ultrafast ion conduction.

Moreover, a temperature-sensitive reduction in the semicircle diameters was observed for three samples as expected (**Figure S3**, Supporting Information). By fitting the EIS data, the Li-ion conductivities were quantitatively deduced as a function of temperature. The results are shown in **Figure 2**d. At 25 ºC, $(LiBH_4)_2 \cdot AB$ delivers a Li-ion conductivity of 1.47×10^{-5} S cm^{-1}, which is nearly three orders of magnitude higher than that of $LiBH_4$ (3.49×10^{-8} S cm^{-1}). More encouragingly, a further increase in the Li-ion conductivity by one order of magnitude was achieved for $LiBH_4 \cdot AB$, reaching 4.04×10^{-4} S cm^{-1}. To the best of our knowledge, this is the highest room-temperature Li-ion conductivity of $LiBH_4$-based SSEs reported so far under identical conditions, which attained a level that is comparable to the widely commercial organic liquid electrolytes (10^{-4} - 10^{-3} S cm^{-1}). At 40 ºC, the Li-ion conductivities of $LiBH_4$, $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$ were determined to be 2.34×10^{-7}, 1.18×10^{-4} and 4.86×10^{-3} S cm^{-1}, respectively. In specific, the incorporation of AB into $LiBH_4$ structure results in a significantly enhanced Li-ion conductivity, especially for the sample with higher AB content, i.e., $LiBH_4 \cdot AB$. However, further increasing the content of AB in $LiBH_4$-xAB from 1:1 to 1:2 induced a decrease in the Li-ion conductivity (**Figure S4**, Supporting Information). It is therefore believed that the optimal composition was 1:1 in the present study. The likelihood is closely related to their crystal structures. As reported previously,[25,26] incorporating AB groups into the $LiBH_4$ structure did not change the basic structural configurations of BH_4, but remarkably enlarged the cell volume. The cell volume of $LiBH_4 \cdot AB$ was determined to be 958.65 Å^3, which is 4 times more than that of $LiBH_4$ (213.519 Å^3) and 1.4 times of that of $(LiBH_4)_2 \cdot AB$ (681.203 Å^3). This effectively decreases the volume density of Li ion, and consequently favors the high Li-ion conduction. In addition, it should be mentioned that the Li-ion conductivity of $LiBH_4$-AB mixtures with molar ratios of 1:1.5 and 1:2 was only lower slightly than that of the 1:1 sample, indicating that the stoichiometric $LiBH_4 \cdot AB$ phase dominated the contribution to the overall conduction. More importantly, the Li-ion conductivity of the $LiBH_4 \cdot AB$ complex is even twice more than that of the complex of $LiBH_4$ with NH_3 (i.e. $LiBH_4 \cdot NH_3$) recently reported (2.21×10^{-3} S cm^{-1}), which provide strong support for our hypothesis above that AB incorporation into $LiBH_4$ is more effective than NH_3 in improving Li-ion conduction due to the larger molecular volume of AB. In addition, no hysteresis was observed upon heating and cooling cycling (18 – 55 ºC) (**Figure S5**, Supporting Information), especially for $LiBH_4 \cdot AB$, as can be seen that the Li ionic conductivity remained still at 4.04×10^{-4} S cm^{-1} at 25 ºC, which means that the ultrafast Li-ion conduction possesses a good stability. This performance is highly desired as practical SSEs.

The direct current (DC) conductivity of $(LiBH_4)_x \cdot AB$ was evaluated by using Li electrodes and SUS electrodes. As shown in **Figure 3**, the DC conductivities of $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$ were determined to be 6.4×10^{-4} S cm^{-1} (50 ºC) and 4.8×10^{-3} S cm^{-1} (40 ºC) for Li electrodes after 1000 s. These values are in good agreement with the results obtained by EIS experiments, suggesting that there exists nearly no obvious impedance between the Li electrodes and electrolyte interfaces. For SUS electrodes, a rapid decrease in the DC conductivity occurred within initial 100 s due to the polarization, and then stabilized at 1.1×10^{-6} S cm^{-1} for $(LiBH_4)_2 \cdot AB$ (50 ºC) and 3.5×10^{-6} S cm^{-1} for $LiBH_4 \cdot AB$ (40 °C), which are nearly two or three orders of magnitude lower than those for Li electrodes. We then calculated the Li-ion transference number (t^+) for $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$ by the following equation.[37-39]

$$t^+ = 1 - \frac{D_{SUS}}{D_{Li}} \qquad (2)$$

where D_{SUS} and D_{Li} are the DC conductivities for SUS electrodes and Li electrodes, respectively. The Li-ion transference number for $(LiBH_4)_2 \cdot AB$ was 0.9982 at 50 ºC and that for $LiBH_4 \cdot AB$ at 40 °C was 0.9993. We therefore believe that the electron conduction is nearly negligible for $(LiBH_4)_x \cdot AB$, that is, the migration of Li ions contributes solely to the charge transference in the present study.

Li Ion Conduction Mechanisms of $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$. To theoretically understand the enhanced Li ionic conduction of $(LiBH_4)_x \cdot AB$, we performed Ab initio molecular dynamics (AIMD) simulations under the Born-Oppenheimer approximation. To speed up diffusion and shorten the simulation time scale, the AIMD simulations were performed at 900 K. **Figure 4** displays the trajectories of Li diffusion in $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$ during AIMD simulations. The simulation results revealed that Li diffusion in $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$ was quite different from the 2D ion conduction occurred in $LiBH_4$.[17] In the $(LiBH_4)_2 \cdot AB$ structure, we observed an apparent 3D diffusion network, consisting of dominating 2D diffusion in the *ac* plane and 1D diffusion channels along the *b* direction (**Figure 4**a-d). This 2D diffusion mode is strongly dependent on the crystal structure of $(LiBH_4)_2 \cdot AB$, which is composed of alternating layers of borohydride and AB (**Figure S2b**, Supporting Information), and the essential structural arrangements of BH_4^- anions and Li^+ cations in the borohydride layers remain almost unchanged compared to pristine $LiBH_4$.[25] Here, the diffusion along the *b* direction should be possible due to the large separation (2.439 Å) between the neighboring molecules in the AB layers. In contrast, the 1D diffusion along the *b* direction dominated in the $LiBH_4 \cdot AB$. This is attributed to the unique structural arrangement of $LiBH_4 \cdot AB$, in which $LiBH_4$ components gather into columns and are separated by AB molecules (**Figure S2c**, Supporting Information).[26] Additionally, we also observed significant Li hopping in the *ac* plane because there is empty space between $LiBH_4$ component and AB molecules. Unlike $(LiBH_4)_2{\cdot}AB$, however, the Li diffusion in the *ac* plane of $LiBH_4{\cdot}AB$ is blocked by AB components, consequently inaccessible to the interchannels, as shown in **Figure 4**g.

The self-diffusion coefficients were further computed from the mean-square displacements (MSDs) using the Einstein relation.[40]

$$D = \frac{1}{6}\lim_{t\to\infty}\frac{d}{dt}\langle (r_i(t) - r_i(0))^2 \rangle \qquad (3)$$

where $r_i(t)$ is the position of the atom i at time t. The slope of the MSD versus time in the normal diffusion region was used to estimate the self-diffusivity in each case. The prefactor of 1/4 was used for the 2D diffusion, and 1/2 for 1D diffusion. The calculated MSDs are plotted in **Figure 5a**. Accordingly, the overall self-diffusion coefficients were determined to be 0.665 and 1.125 $Å^2$ ps^{-1} for $(LiBH_4)_2{\cdot}AB$ and $LiBH_4{\cdot}AB$, respectively. It is clear that the Li diffusion in $LiBH_4{\cdot}AB$ is remarkably faster than that in $(LiBH_4)_2{\cdot}AB$, possibly because of the enlarged cell volume caused by the incorporation of more AB groups. This is also in excellent agreement with our experimental results (**Figure 2**). Moreover, the detailed Li-diffusion process was understood by calculating and comparing the diffusion coefficients along *a*, *b*, *c* directions and in *ab*, *bc*, *ac* planes (**Table 1**). For $(LiBH_4)_2{\cdot}AB$, the Li-ion diffusion along the *a* and *c* directions is faster than along the *b* direction, further confirming a favorable diffusion in the *ac* plane (0.708 $Å^2$ ps^{-1}). In particular, the fastest Li diffusion along the *a* direction should be the diffusion majority in the *ac* plane. In the $LiBH_4{\cdot}AB$ structure, the Li diffusion along the *c* direction seems to be the fastest, possibly due to the presence of empty space between $LiBH_4$ component and AB molecules, which subsequently makes Li hopping much easier from $LiBH_4$ to the empty space. Unfortunately, this diffusion pathway is blocked by AB molecules. As a result, the diffusion along the *b* direction should dominate the ion conduction of $LiBH_4 \cdot AB$ with a diffusion coefficient of 1.061 $Å^2\ ps^{-1}$.

Further calculations and comparison in the diffusion energy barriers were also conducted. Shown in **Figure S6** (Supporting Information) are the optimized diffusion pathways. As expected, the diffusion in the *ac* planes and along the *b* direction dominated the overall Li diffusion in $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$, respectively. **Figure 5**b illustrates quantitatively the activation energy barriers of $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$. The activation energy value for $LiBH_4 \cdot AB$ was estimated to be only 0.12 eV, half of that for $(LiBH_4)_2 \cdot AB$ (0.25 eV), representing a much more favorable diffusion in the $LiBH_4 \cdot AB$ structure. Moreover, it is noteworthy that these activation energy values (0.25 eV for $(LiBH_4)_2 \cdot AB$ and 0.12 eV for $LiBH_4 \cdot AB$) are substantially lower than the reported typical activation energies for super ion conductors (≈ 0.5 eV),[41] which may be responsible for the ultrafast Li-ion conductivity of lithium borohydride ammonia borane complexes at room temperature in the present study.

Electrochemical Stability of $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$. The electrochemical stability of $(LiBH_4)_x \cdot AB$ was evaluated by LSV and CV measurements. As shown in **Figure 6**, the LSV results revealed the oxidative stability limit of 2.45 V for $(LiBH_4)_2 \cdot AB$ and 2.04 V for $LiBH_4 \cdot AB$. The stripping/plating of Li was also visible in the CV curves between -0.5 V and 0.5 V at a scan rate of 10 $\mu V\ s^{-1}$ (**Figure S7,** Supporting Information), and there was no appreciable current peak at 0.5-1.5 V caused by the decomposition of electrolyte. We therefore believe that $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$ remained stable used as SSE in the potential range from 0 to 2 V (vs Li^+/Li), similar to the $LiBH_4$ SSE reported previously.[42,43] Additionally, galvanostatic cycling was conducted to evaluate the chemical stability of $(LiBH_4)_x \cdot AB$ electrolyte when paired with Li metal in $Li|(LiBH_4)_x \cdot AB|Li$ cells. **Figure 7**a and b present the typical voltage profiles as a function of time at a constant density of 0.2 mA cm^{-2}. The flat voltage plateaus with 50 mV of overpotential were observed in the voltage profiles of $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$. Moreover, no apparent voltage fluctuations were detected after cycling for 40 h, suggesting a good long-term cycling stability. **Figure 7**c and d show the voltage profiles with elevated current density. Clearly, elevating the working current density induces an increase in the overpotential at 0.1-3.0 mA cm^{-2}. An abrupt increase in the overpotential occurred at 1.8 and 3.0 mA cm^{-2}, possibly due to the formation of Li dendrite, for $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$, respectively. Here, it is worth highlighting that the critical current density of 3.0 mA cm^{-2} for $LiBH_4 \cdot AB$ is higher than those for most of reported SSE materials under identical conditions. Thus the $(LiBH_4)_x \cdot AB$ complexes possess both good chemical stability and high anti-Li-dendrite growth of Li, which is very favorable as a potential SSE material.

CONCLUSIONS

Two lithium borohydride ammonia borane complexes, $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$ were prepared. Both $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$ exhibited ultrafast room-temperature Li-ion conduction, thanks to the decreased volume density of Li ions caused by the enlarged cell volume. In particular, $LiBH_4 \cdot AB$ delivered a Li-ion conductivity of 4.04×10^{-4} S cm^{-1} at 25 ºC. Such ion conductivity is the highest value among all so far reported $LiBH_4$-based SSEs under identical conditions, and comparable to the widely used commercial organic liquid electrolytes. The ultrafast room-temperature Li-ion conductivity exhibited good stability upon heating and cooling cycling. Moreover, nearly 0.999 Li ion transference numbers were obtained for both complexes, indicating negligible electron conduction. AIMD simulations revealed that $(LiBH_4)_2 \cdot AB$ is a 3D ion conductor with 2D diffusion in the *ac* plane and 1D diffusion channels along the *b* direction, while $LiBH_4 \cdot AB$ is a 1D ion conductor with the fast diffusion along *b* direction within the $LiBH_4$ columns. The various diffusion modes were structure dependent. Moreover, the activation energy barrier value for $LiBH_4{\cdot}AB$ was 0.12 eV, only half of that for $(LiBH_4)_2{\cdot}AB$ (0.25 eV), which well explains the much favorable diffusivity in the $LiBH_4{\cdot}AB$ structure. The electrochemical working potential window was determined to be 0 - 2.0 V for $(LiBH_4)_2{\cdot}AB$ and $LiBH_4{\cdot}AB$ by CV measurements. Moreover, the critical current density of the formation of Li dendrite for $LiBH_4{\cdot}AB$ was estimated to be 3.0 mA cm^{-2}, which is higher than those of reported SSE materials under identical conditions. Such good chemical stability and high anti-dendrite-growth property of Li are very favorable as potential SSE materials.

ASSOCIATED CONTENT

Supporting Information: The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/

XRD profiles of $LiBH_4$-xAB mixtures, Crystal structures, EIS spectra at different temperatures, ionic conductivity plots upon heating/cooling cycles, ionic conductivity plots of $LiBH_4$-xAB mixtures, Li diffusion pathways, and CV curves of $(LiBH_4)_x{\cdot}AB$ (PDF)

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NOTES

The authors declare no competing financial interest.

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Table 1. Calculated diffusion coefficients along *a*, *b*, *c* directions and in *ab*, *ac*, *bc* planes.

Samples	Diffusion coefficients (Å^2 ps^{-1})						
	a	b	c	ab	ac	bc	abc
$(LiBH_4)_2$·AB	0.824	0.578	0.593	0.701	0.708	0.585	0.665
$LiBH_4$·AB	0.924	1.061	1.390	0.992	1.157	1.226	1.125

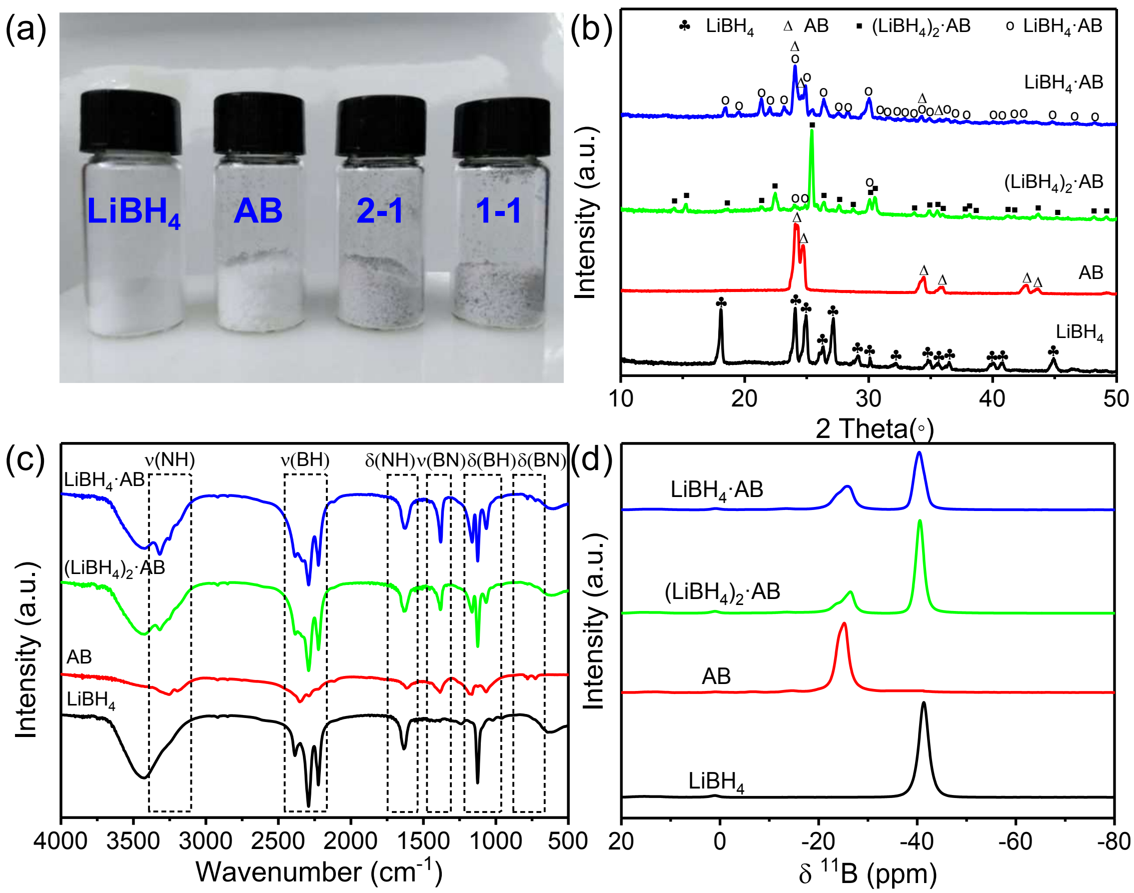

Figure 1. (a) Photograph, (b) XRD patterns, (c) FTIR spectra and (d) solid-state ^{11}B NMR spectra for $LiBH_4$, AB, $(LiBH_4)_2$·AB and $LiBH_4$·AB.

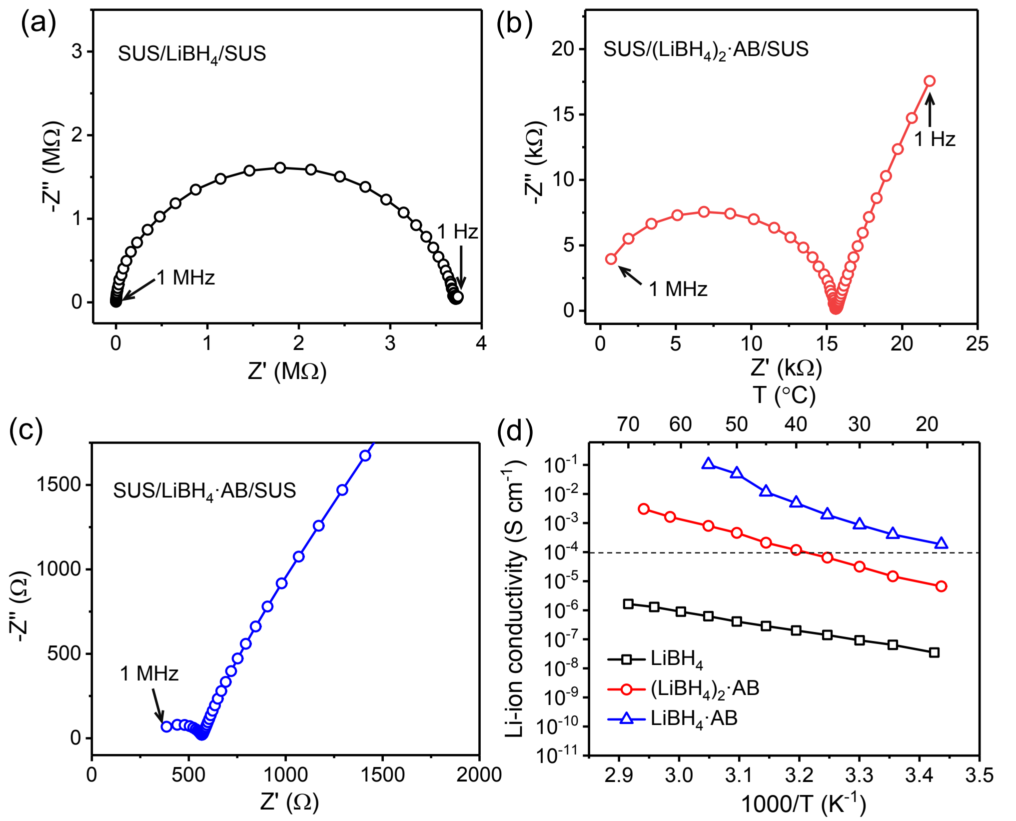

Figure 2. Nyquist (a-c) and Arrhenius ionic conductivity (d) plots of $LiBH_4$, $(LiBH_4)_2{\cdot}AB$ and $LiBH_4{\cdot}AB$.

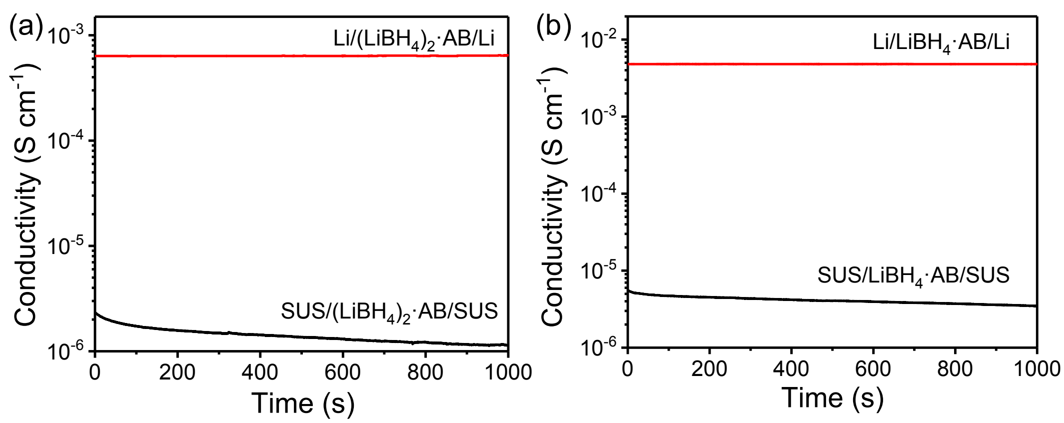

Figure 3. Time dependence of DC conductivity of (a) $(LiBH_4)_2{\cdot}AB$ at 50 ºC and (b) $LiBH_4{\cdot}AB$ at 40 ºC.

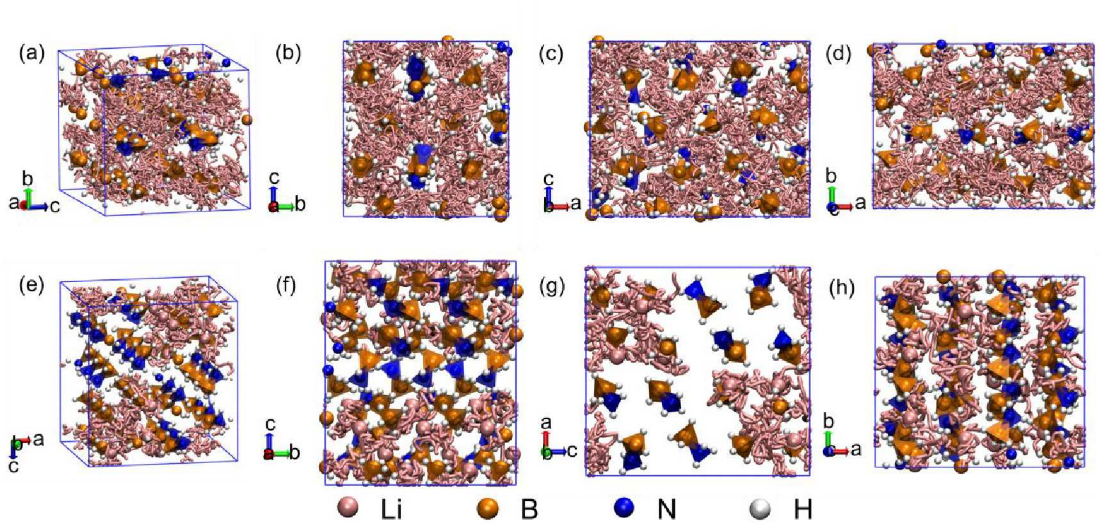

Figure 4. Trajectories (pink) of Li ions obtained from Ab initio molecular dynamics simulations at 900 K. The Li-ion diffusion pathways for a-d) $(LiBH_4)_2 \cdot AB$ and e-h) $LiBH_4 \cdot AB$ are shown in different views.

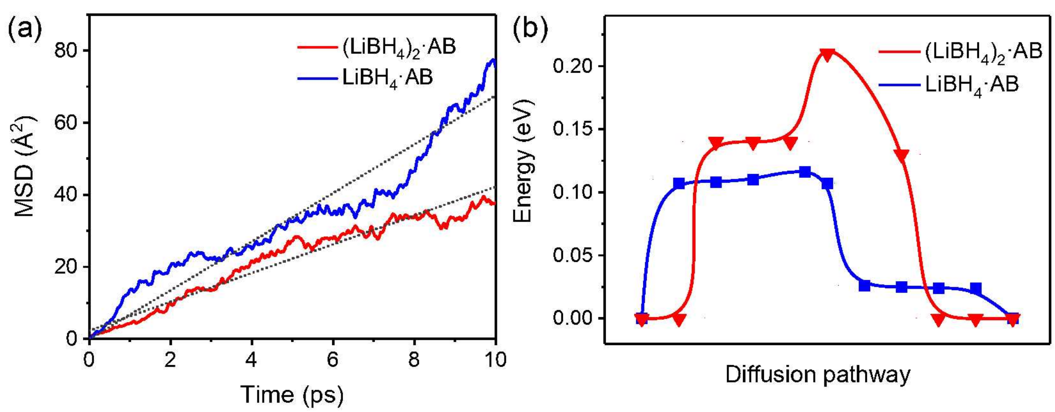

Figure 5. (a) Mean square displacement and (b) activation energy barriers of $(LiBH_4)_2 \cdot AB$ and $LiBH_4 \cdot AB$.

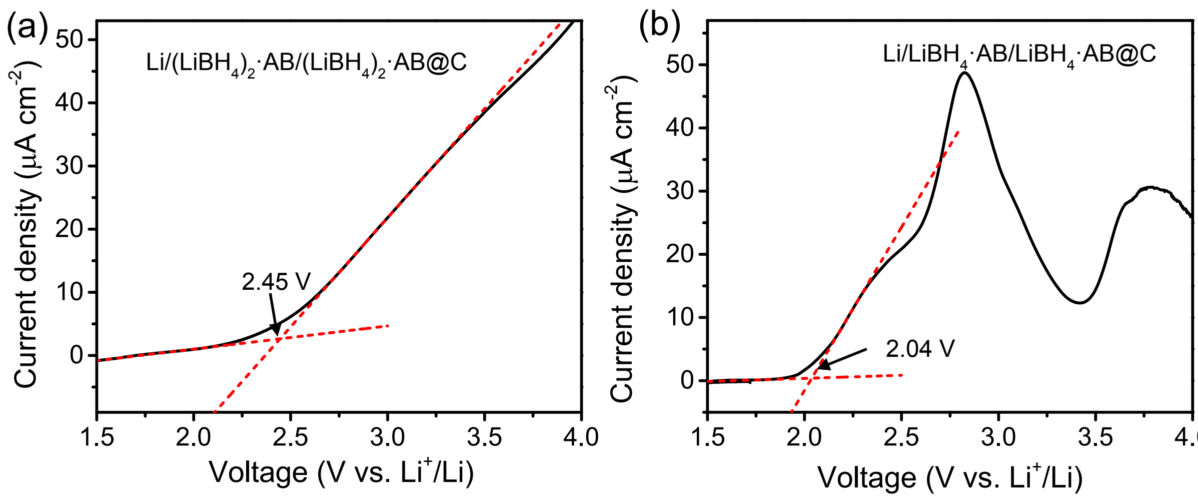

Figure 6. LSV curves of (a) $(LiBH_4)_2$·AB at 30 °C and (b) $LiBH_4$·AB at 25 °C between 1.5 and 4.0 V at a scan rate of 10 μV s^{-1}.

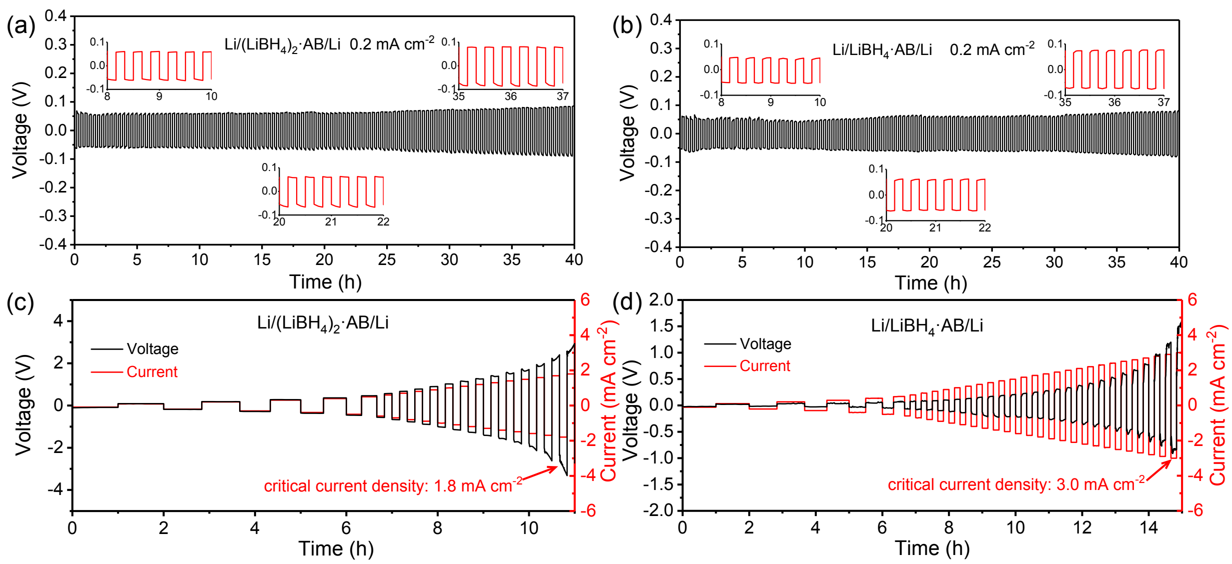

Figure 7. Constant current GC of (a) Li/$(LiBH_4)_2$·AB/Li cell at 50 ºC and (b) Li/$LiBH_4$·AB/Li cell at 35 ºC with a current density of 0.2 mA cm^{-2}. Stepped current density GC for (c) Li/$(LiBH_4)_2$·AB/Li cell at 50 ºC and (d) Li/ $LiBH_4$·AB/Li cell at 35 ºC. The inset profiles show the detailed voltage plateau of Li stripping/plating.

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